

# Tahsin Tajwar Tanni

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[Portfolio](#) | [LinkedIn](#) | [Github](#) | [Google Scholar](#)

## PERSONAL PROFILE

Computer Science and Engineering graduate with a thesis on pretrained language model security, focused on defense mechanisms against weight noise and low precision quantization attacks. Skilled in deep learning research with PyTorch and Hugging Face Transformers, LLM application engineering with the OpenAI and Anthropic APIs including RAG and agentic workflows, and reproducible, statistically grounded empirical analysis in Python

## EDUCATION

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|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 01/2022 – 03/2026 | <b>BRAC UNIVERSITY</b><br><b>BSc in Computer Science and Engineering</b><br>Relevant Coursework: Artificial Intelligence, Neural Networks | GPA: 3.60 / 4.00 |
| 03/2018 – 11/2020 | <i>Govt. Haji Muhammad Mohsin College</i><br><b>Higher Secondary Certificate (HSC)</b>                                                    | GPA: 5.00 / 5.00 |
| 01/2004 – 12/2018 | <i>Bangladesh Mohila Samity Girls' High School and College</i><br><b>Secondary School Certificate (SSC)</b>                               | GPA: 5.00 / 5.00 |

## RESEARCH EXPERIENCE

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|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9/2026 – Present  | <b>Student Researcher</b><br><i>Elite Research Lab LLC</i> <ul style="list-style-type: none"><li>• Lead a four-person team on Relational Boundary Collapse, a construct and turn-level detection framework for unhealthy relationship formation in mental health chatbot conversations; proposal selected through the lab's Research Training and Competition 2026.</li></ul>                                                                                                                                                                         |
| 01/2025 – 01/2026 | <b>Undergraduate Researcher</b><br><i>BRAC University, Department of Computer Science and Engineering</i> <ul style="list-style-type: none"><li>• Built the evaluation dataset and implemented the defense for Anchor-Guided Repair, a fine-tuning method that anchors attention and MLP projections toward a frozen clean copy to repair models damaged by weight noise and low-precision quantization.</li><li>• Recovered 60 to 80 percent of attack-induced loss on Qwen2.5-0.5B and Pythia-410M across FP16, INT8 and 4-bit precision.</li></ul> |

## PUBLICATIONS

### *Published*

Rohan, A. M., Khan, N., **Tanni, T. T.**, Fardin, F., & Bushra, A. (2026). *Anchor-Guided Repair: A Defense Mechanism for Enhancing Stability of Compromised Pretrained Language Models Against Low-Precision and Weight Noise Attacks*. In *Proceedings of IndabaX Nigeria 2026: Building Scalable AI That Works: From Research to Deployment in Resource-Constrained Environments. Proceedings of Machine Learning Research*, **319**, 368–381.  
<https://proceedings.mlr.press/v319/rohan26a.html>

### *Under Review*

**Tanni, T. T.**, & Khan, N. *Reporter Coverage and Campaign Duplication in GHSA/OSV Malware Advisories*. [Manuscript submitted for publication]. MSR 2027.

- Showed 994 GHSA/OSV malware advisories collapse to 317 independent campaigns, and that text-derived labels depend on the reporting analyst (OR  $\approx$  23).
- Found a well-powered null for package naming vs. payload mechanism; labeler validated by two annotators ( $\kappa$  0.78/0.66); code and data released.

## PROJECTS

### **RAG-Slackbot**

- Built a Slack-integrated, tool-augmented LLM agent enabling real-time querying of internal knowledge bases, including automated document ingestion, chunking, vector storage, and context-aware generation. Implemented confidence-based escalation and

source attribution to constrain the agent's autonomous actions when retrieved context was insufficient or unreliable.

- Tools Used: n8n, Supabase, Ollama, Google Gemini, Slack API

## ThesisFlow

- Designed and built a role-based thesis management platform handling submission, tracking, and supervision workflows across student, supervisor, and administrator roles. Recognized with 2nd Runner-up at the CSE471 Project Demonstration

- Tools Used: MongoDB, Express.js, React, Node.js

## PROFESSIONAL EXPERIENCE

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05/2026 – Present      **AI Automation Engineer**

*180 RE, Toronto, Canada*

- Build LLM-powered workflows using n8n, automating multi-step business processes via n8n and third-party service integrations. Develop AI-powered chatbots, integrating them into websites to automate user interactions and supports

02/2023 – 07/2023      **Junior Web Developer**

*Weabers, Chattogram, Bangladesh*

- Built responsive web applications; collaborated on API integration and feature deployment.

## VOLUNTEERING

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02/2025 - 07/2025      **Python Instructor**

*Zentorra*

- Taught foundational Python programming concepts to beginner-level learners.

## TECHNICAL SKILLS

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- **Programming Languages:** Python, JavaScript, C
- **Machine Learning & AI:** PyTorch, Hugging Face Transformers, scikit-learn, Deep Learning, Natural Language Processing (NLP)
- **Research Methods:** Model Fine-tuning, Transfer Learning, Adversarial Machine Learning, Performance Benchmarking, Reproducible Research
- **Scientific Computing:** NumPy, pandas
- **Tools & Platforms:** Docker, Git, GitHub, Hugging Face Spaces, LaTeX
- **Software Development:** React.js, Node.js, Express.js, MongoDB, MySQL, SQLite

## LANGUAGE PROFICIENCY

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- **English:** Full professional proficiency - Medium of Instruction (MOI) certified, BRAC University, 2026
- **Bengali:** Native

## AWARDS & HONORS

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- **Dean's List, BRAC University**, Fall 2024
- **Vice Chancellor's List, BRAC University**, Summer 2025
- **2nd Runner-up - CSE471 Project Demonstration**, BRAC University, 2025
- **Academic Distinction**, BSc in Computer Science and Engineering, BRAC University, 2026

## REFERENCES

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**Dr. Muhammad Iqbal Hossain**

*Thesis Supervisor*

Associate Professor,

BRAC University

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